PART A

Differentiate drift and diffusion movement of carriers in semiconductors. (3)

4 Derive continuity equation. (3)

3)

Drift and Diffusion Currents

- **Current**
  Generated by the movement of charged particles (negatively charged electrons and positively charged holes).

- **Carriers**
  The charged electrons and holes are referred to as **carriers**

- The two basic processes which cause electrons and holes move in a semiconductor:
  - Drift - the movement caused by electric field.
  - Diffusion - the flow caused by variations in the concentration.

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3)

Drift Current Vs Diffusion Current

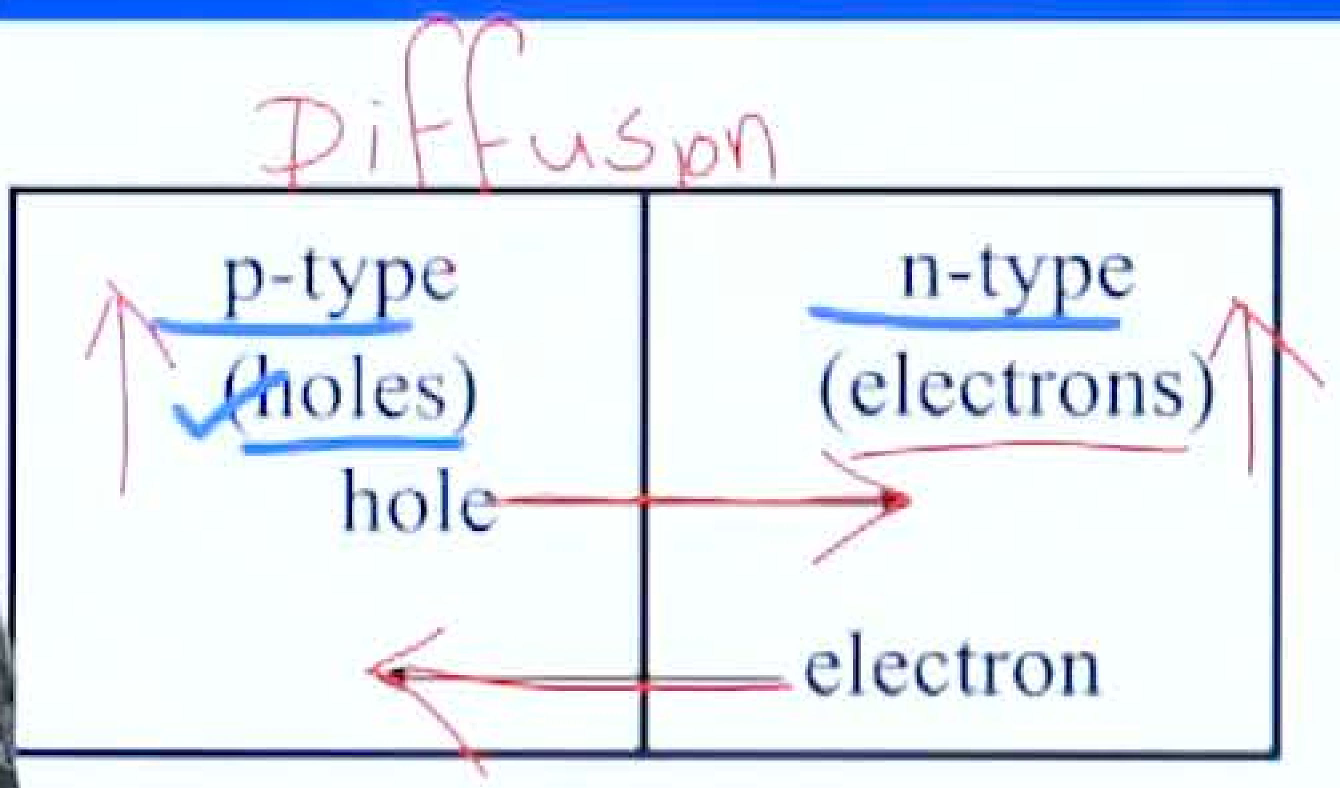

Figure - Diffusion Current

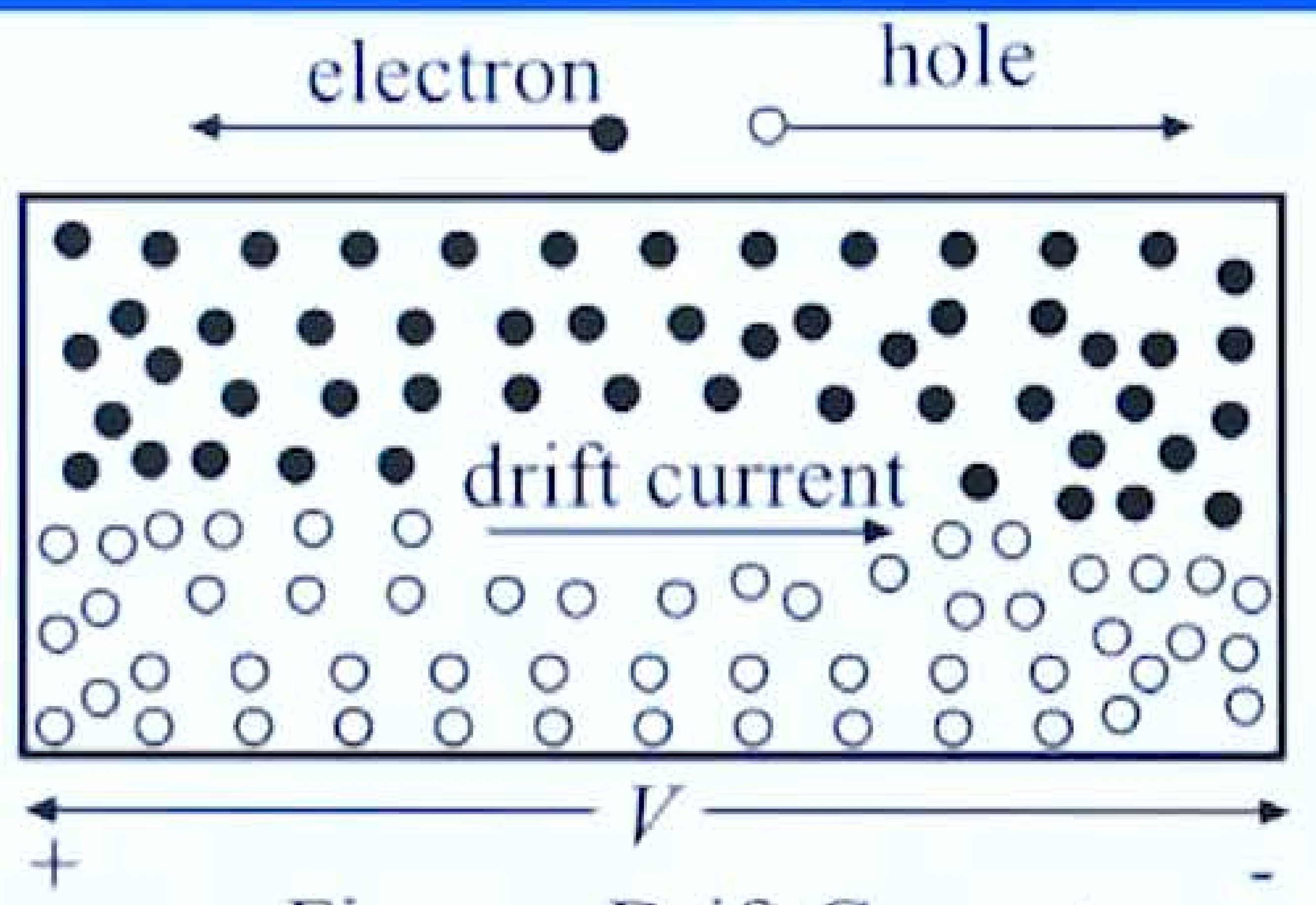

Figure - Drift Current

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PART A

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4 Derive continuity equation. (3)

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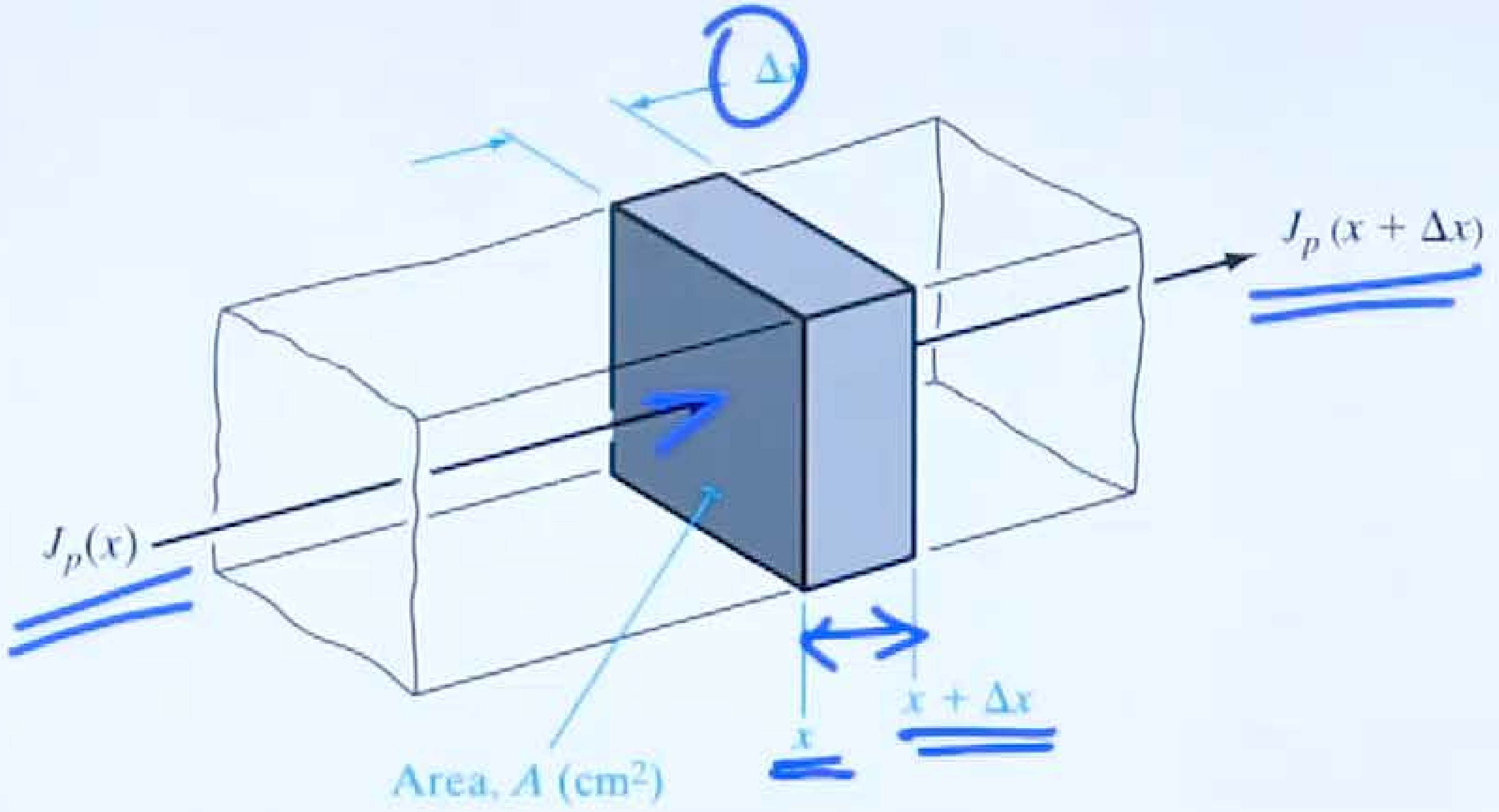
4)



Current entering and leaving a volume $\Delta x A$.

$$\left.\frac{\partial p}{\partial t}\right|_{x \to x+\Delta x} = \frac{1}{q}\frac{J_p(x) - J_p(x+\Delta x)}{\Delta x} - \frac{\delta p}{\tau_p}$$

Rate of hole buildup = increase of hole concentration in $\delta x A$ per unit time − recombination rate

As Δx approaches zero, we can write the current change in derivative form:

$P = P_0 + \delta P$

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p}$$

The expression (4–31a) is called the *continuity equation* for holes. For electrons we can write

$$\frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n}$$

PART B

13 (a) State and prove Einstein's relation. (9)

(b) An n type silicon bar 0.1 cm long and 100 μm^2 in cross sectional area has a majority carrier concentration of $5 x 10^{15}$ cm^{-3} and electron mobility μ_n=1300 cm^2/Vs at 300K. What is the resistance of the bar? (5)

14 (a) Explain Hall Effect? Derive the expression for carrier concentration and mobility in terms of Hall voltage. (7)

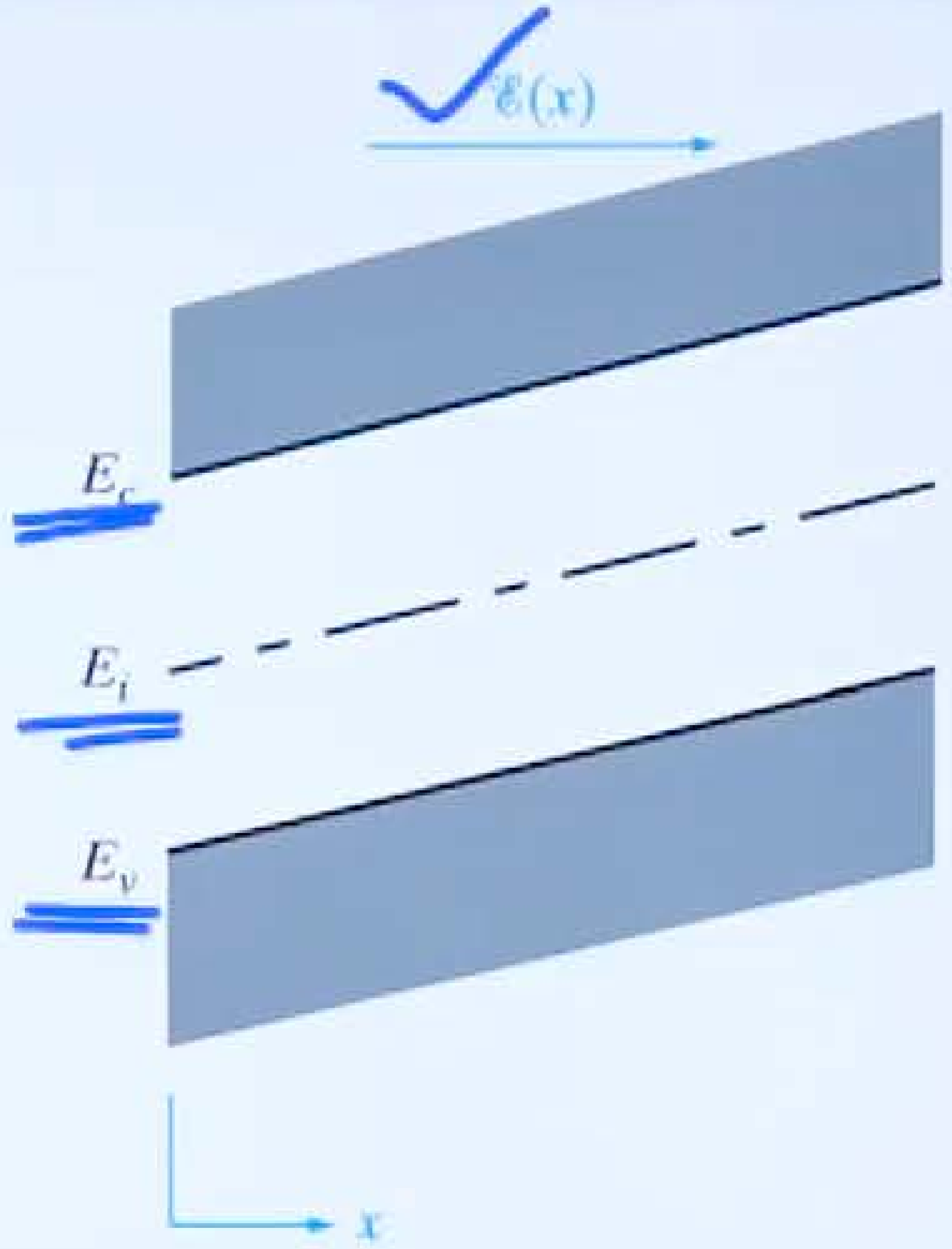
(b) Derive the expression for diffusion current density in a semiconductor. (7)

Pause

ONES

13 a)

Energy band diagram of a semiconductor in an electric field $\mathcal{E}(x)$.



From the definition of electric field,

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx}$$

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$$\mathcal{E}(x) = -\frac{dV(x)}{dx} = -\frac{d}{dx}\left[\frac{E_i}{(-q)}\right] = \frac{1}{q}\frac{dE_i}{dx} \quad -(1)$$

Drift

$$J_p(x) = q\mu_p p(x)\mathcal{E}(x) - qD_p\frac{dp(x)}{dx} = 0$$

Equating to zero at equilibrium

$$\mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{p(x)}\frac{dp(x)}{dx}$$

Pause

Using

$$p_0 = n_i e^{(E_i - E_F)/kT}$$

$$\mathscr{E}(x) = \frac{D_p}{\mu_p}\frac{1}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right)$$

$\to 0$

$$\frac{D_p}{\mu_p} \cdot \frac{1}{KT} \cdot q\, \cancel{E(x)}$$

On simplification

$$\frac{D_p}{\mu_p} \cdot \frac{1}{KT} \cdot q = 1$$

$$\boxed{\frac{D}{\mu} = \frac{kT}{q}}$$

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PART B

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Pause

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13 b)

Resistivity

~~Conductivity~~ $\rho=1/(qn_0\mu_n)$. $= \frac{1}{\sigma}$

Resistance of the bar, $R = \rho L/A$.

$L = 0.1\,cm$

$A = 100\,\mu m^2$

$\sigma = n_0 q \mu_n$

$\rho = \frac{1}{\sigma} = \frac{1}{n_0 q \mu_n}$

$n_0 = 5 \times 10^{15}$

$q = 1.6 \times 10^{-19}$

$\mu_n = 1300$

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PART B

13 (a) State and prove Einstein's relation. $\frac{D}{\ } = \frac{KT}{\ } = V_T$ (9)

(b) An n type silicon bar 0.1 cm long and 100 μm^2 in cross sectional area has a majority carrier concentration of 5×10^{15} cm^{-3} and electron mobility $\mu_n = 1300$ cm^2/Vs at 300K. What is the resistance of the bar? (5)

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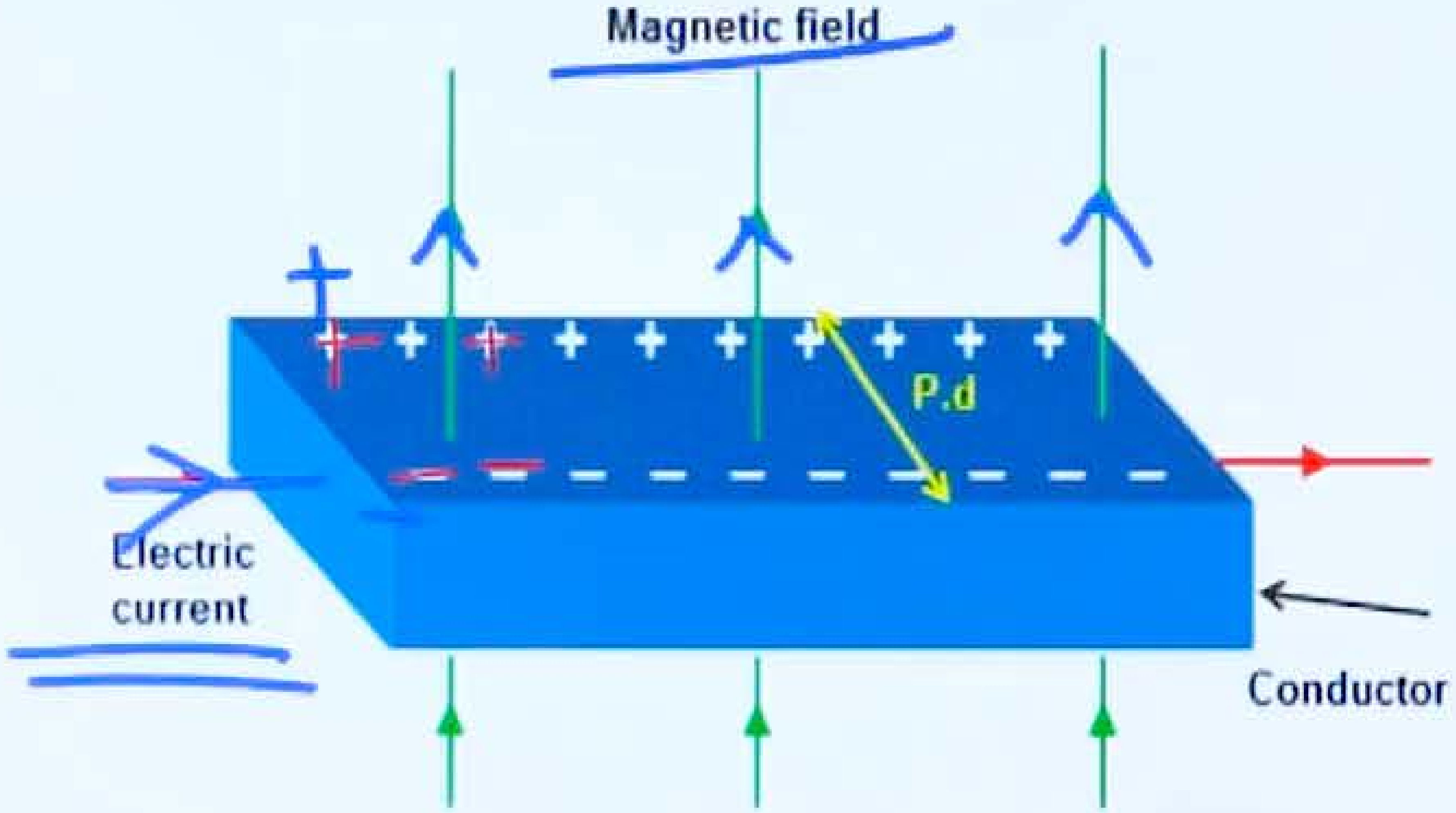
14 a) When a magnetic field is applied to a current carrying conductor in a direction perpendicular to that of the flow of current, a potential difference or transverse electric field is created across a conductor. This phenomenon is known as Hall Effect.



P.d = Potential difference

Hall Effect

Under equilibrium conditions , the force on the charges due to HALL electric field and LORENTZ force counterbalances each other, that is

$$\cancel{q}E_y = \cancel{q}\, v_x B_z \quad \text{-------------} \; (1)$$

$$E_y = v_x B_z \quad \text{-------------} \; (2)$$

We know that the CURRENT DENSITY 'J_x' is given by

$$J_x = p\,q\,v_x \quad \text{------------} \; (3)$$

$$v_x = \frac{J_x}{pq} \quad \text{------------} \; (4)$$

$I \rightarrow x \quad v_x$

$$\frac{1}{pq} = R_H$$

$$E_y = \frac{J_x}{pq\,\cancel{B_z}} \cdot B_z = J_x R_H B_z$$

Where R_H is Hall coefficient.

$$R_H = \frac{1}{pq} \quad ----$$

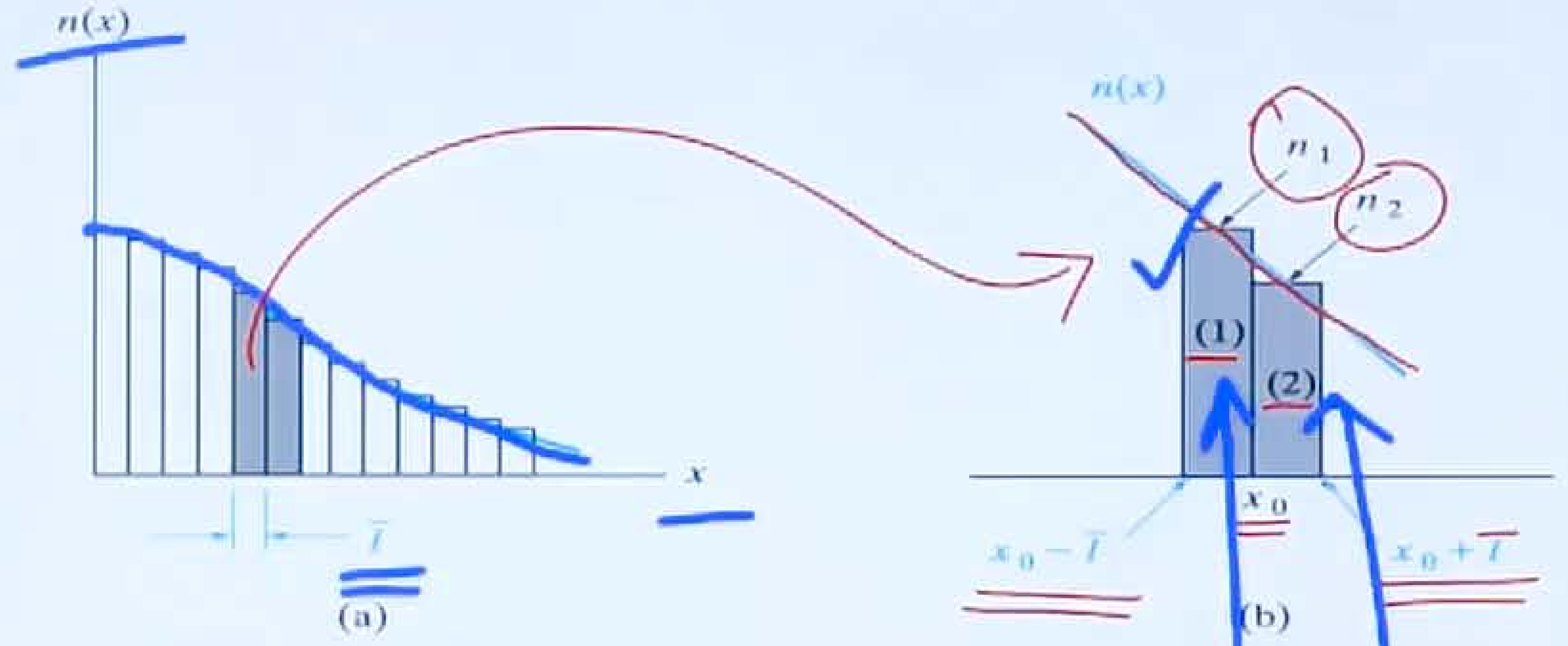
$$P = \frac{1}{R_H \cdot q}$$

Conductivity, $\sigma = pq\mu_p$

$$\mu_p = \frac{\sigma}{pq} = \sigma \cdot R_H = \frac{R_H}{\rho}$$

$$\mu_p = \sigma R_H = \frac{R_H}{\rho}$$

14 b)



The electrons in segment (1) to the left of x_0 in Fig. 4–13b have equal chances of moving left or right, and in a mean free time $\bar{t}$ one-half of them will move into segment (2). The same is true of electrons within one mean free path of x_0 to the right; one-half of these electrons will move through x_0 from right to left in a mean free time. Therefore, the *net* number of electrons passing x_0 from left to right in one mean free time is $\frac{1}{2}(n_1 \bar{l} A) - \frac{1}{2}(n_2 \bar{l} A)$, where the area perpendicular to x is A. The rate of electron flow in the $+x$-direction per unit area (the electron flux density ϕ_n) is given by

$$\phi_n(x_0) = \frac{\bar{l}}{2\bar{t}}(n_1 - n_2) \tag{4–18}$$

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ONES

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Since the mean free path $\bar{l}$ is a small differential length, the difference in electron concentration $(n_1 - n_2)$ can be written as

$$n_1 - n_2 = \frac{n(x) - n(x + \Delta x)}{\Delta x}\bar{l} \tag{4–19}$$

where x is taken at the center of segment (1) and $\Delta x = \bar{l}$. In the limit of small Δx (i.e., small mean free path $\bar{l}$ between scattering collisions), Eq. (4–18) can be written in terms of the carrier gradient $dn(x)/dx$:

$$\phi_n(x) = \frac{\bar{l}^2}{2\bar{t}} \lim_{\Delta x \to 0} \frac{n(x) - n(x + \Delta x)}{\Delta x} = \frac{-\bar{l}^2}{2\bar{t}} \frac{dn(x)}{dx} \tag{4–20}$$

Diffusion Coefficient

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The diffusion current crossing a unit area (the current density) is the particle flux density multiplied by the charge of the carrier:

$$J_n(\text{diff.}) = -(-q)D_n\frac{dn(x)}{dx} = +qD_n\frac{dn(x)}{dx} \quad (4\text{–}22\text{a})$$

$$J_p(\text{diff.}) = -(+q)D_p\frac{dp(x)}{dx} = -qD_p\frac{dp(x)}{dx} \quad (4\text{–}22\text{b})$$